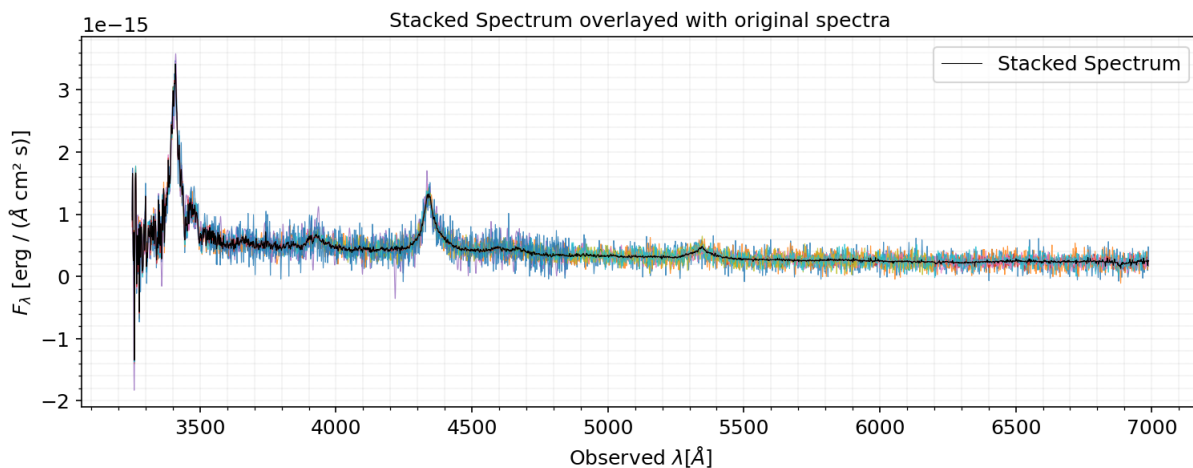


easyspec News

Stacking spectra collected with different instruments

The new `easyspec` function `analysis.stack_calib_spectra()` allows the user to stack spectra of the same target collected with different instruments and configurations. In the image below, we show the stacked spectrum (black line) of a blazar, obtained by combining 16 spectra across different wavelength ranges and spectral resolutions (colored, noisy background spectra).



The available stacking methods are "mean" and "median", and the final spectral resolution is increased by a factor > 1 (set by the user) with respect to the spectrum with the best spectral resolution. Note that this increased spectral resolution results from interpolation. The real power of this method lies in improving F_λ .

How does it work?

Let's suppose you have:

- one spectrum covering the wavelength range from 3500 Å up to 5000 Å.
- a second spectrum from 4000 Å up to 6000 Å.
- a third spectrum from 4000 Å up to 7000 Å.

In the region below 4000 Å, where we have only one spectrum, nothing happens, and the final result is simply the original single spectrum. In the region from 4000 Å to 5000 Å, where all spectra overlap, the stacking will be performed in all three of them. In the range from 5000 Å to 6000 Å, we have two overlapping spectra, so these two will be stacked in this wavelength range. Finally, in the range from 6000 Å to 7000 Å, where we have only one spectrum, nothing happens, and, once more, the final result is simply the original single spectrum.

A **tutorial** for this method is available **on GitHub**:



<https://github.com/ranieremenezes/easyspec/tree/main>